

SURYA BALAJI

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Education

The University of Texas at Austin

May 2028

B.S., Electrical and Computer Engineering Honors, B.B.A., Canfield Business Honors

GPA: 3.96

Relevant Coursework: Real-Time Operating Systems, Embedded Systems, Linear Systems & Signals, Computer Architecture, Linear Algebra, Multivariable Calculus, Differential Equations, Digital Logic Design, Circuit Theory

Relevant Skills

Languages/Libraries: Python, C, C++, ARM M0+ Assembly, TensorFlow, scikit-learn, NumPy, Pandas, OpenCV

Systems/Platforms: ROS 2, Isaac Sim, cuRobo, Docker, Linux, FreeRTOS, Distributed Systems, Real-Time Systems

Tools: Git, GitHub, VSCode, ESP-IDF, Code Composer Studio, Vivado, KiCad, SPICE, IdeaMaker, Autodesk Fusion

Experience

Autonomous Mobile Robotics Laboratory

May 2026 – Present

Undergraduate Researcher

Austin, TX

- Developing perception-driven motion planning on Linux and ROS 2 stack for 7-DOF Kinova robotic arm on mobile base
- Implementing GPU-accelerated, collision-free motion planning via global MotionGen trajectory optimizer with reactive MPC controller in cuRobo v2 against voxelized ESDF environment fused from 3D LiDAR/depth-camera point clouds
- Preparing deployment of autonomous system on NVIDIA Jetson AGX Thor with validated 17 ms trajectory generation latency and pending benchmarks for control loop frequency and collision avoidance accuracy KPIs

The University of Texas at Austin

Aug 2025 – Present

Teaching Assistant, ECE 319K/H: Intro to Embedded Systems

Austin, TX

- Evaluate lab competency across 3 lab sections covering GPIO, FSM design, interrupts, and SPI/ADC/UART drivers
- Hold 3 hrs/week office hours for 320 students to reinforce embedded systems concepts and debug lab implementations
- Collaborate with faculty weekly to diagnose knowledge gaps, write/proctor exams, and align instruction with curriculum

Longhorn Neurotech

Aug 2024 – May 2025

EMG Director & VR HSI Subsystem Lead

Austin, TX

- Enabled wireless peripheral control by thresholding rectified muscle-signal voltage into binary flex-activated commands
- Drove omnidirectional rover by transmitting JSON commands from muscle-signal control pipeline over UDP/Wi-Fi
- Controlled VR character with muscle-signal pipeline by routing commands to Unity environment via FastAPI backend

Projects

Autonomous Racecar & Real-Time Operating System · 1st place out of 12 teams

Jan 2026 – May 2026

- Built and raced custom autonomous Ackermann-differential drive hybrid racecar featuring PD motor controller, dual-microcontroller CAN communication, and Wi-Fi server data logging and start/stop race status acquisition
- Navigated racecar using <1 us jitter multi-rate sensor fusion of LiDAR, IR, and IMU concurrent RTOS threads
- Ran racecar on custom preemptive RTOS kernel with 8-level priority scheduling, PendSV context switching, blocking semaphores, SVC process loader, and FAT file system to enable low-latency multitasking and diagnostic data logging

sEMG Gesture Classification & Prosthetic Arm

Dec 2025 – Present

- Recorded forearm muscle signals from EMG sensors for prosthetic arm control leveraging DMA for 1 kHz non-blocking ADC sampling, asynchronous USB data transfer, onset detection for gesture labeling, and HDF5 training data storage
- Trained LDA and MLP models on features from collected muscle signal data to recognize 5 hand gestures with 82% accuracy via scikit-learn and TensorFlow, models deployed on ESP32 for fast, offline TinyML prosthetic arm control
- Customized 7-DOF prosthetic arm, adding 3D-printed servo supports and modified servo for continuous bicep actuation

The M0+rix: Segfaulted - Real-Time Game Engine · 1st place out of 300+ students

March 2025 – May 2025

- Designed case and KiCad PCB for embedded video game console with LCD screen, DAC audio, and ADC/GPIO input
- Programmed 20 FPS 3D graphics engine in bare-metal C utilizing interrupts, SPI, and 2-player UART communication
- Built asset processing pipeline to convert image and audio data to fit under 128KB flash and 32KB RAM constraints